Lesson notes · for you

Train your own model: unit check

AI · Level 3 · Unit check

Time
45 min

Materials
Paper and a pen. No device needed.

Learning goals

- Data first
- How a classifier decides
- Test it honestly
- Fair for everyone

Activity: Team quiz

1. Make teams of three or four, or quiz a family member
2. Read each question aloud; agree on one answer
3. Reveal the answer and explain why it is right
4. Write the hardest question in your notebook to revise

Discussion

- Which idea from this unit surprised you most?
- What would you check first before trusting an AI tool's score?
- What model would you like to build, and whose data would it need?

Check yourself

- One question I will ask about the next AI tool I use

Try it this week
Pick one AI tool you use (voice typing, a translator, a photo app) and ask the three questions from this unit: what data did it learn from, how was it tested, and who might it fail?

Worksheet

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Name _____ Class _____ Date _____

1 **You want a model that tells healthy from sick cassava leaves. Which dataset is best?**

- ◯ 100 healthy photos and 5 sick ones
- ◯ 60 healthy and 60 sick, all from one plant at noon
- ◯ 30 healthy and 30 sick, from different fields, times and phones
- ◯ Any leaf photos from the internet, unlabelled

2 **A model treats every label in its training data as correct, even the wrong ones.**

◯ True ◯ False

3 **Match each word to its meaning.**

Feature	• • • •	More mistakes for one group than for others
Label	• • • •	Examples kept apart to measure the model honestly
Test set	• • • •	The right answer given to an example
Bias	• • • •	A number you can measure about every example

4 **A new mango's nearest neighbour is ripe, but its next four neighbours are all unripe. What does the classifier say with k = 1, and with k = 5?**

- ◯ Ripe with k = 1, not ripe with k = 5
- ◯ Ripe both times
- ◯ Not ripe both times
- ◯ It can't decide with k = 5

5 **Put the steps of building and checking a model in order.**

- ☐ Test on the test set, group by group
- ☐ Train on most examples, keeping a test set apart
- ☐ Label them and check the labels
- ☐ Collect examples fairly, with permission
- ☐ Plan the data: what, labels, how many, how varied